C U R R I C U L U M V I T A E



Personal Details

Name	**Lukas Folkman**
Address	CeMM, Lazarettgasse 14, AKH BT 25.3, A-1090 Vienna, Austria
E-mail	LFolkman@cemm.oeaw.ac.at
Website	lukas-folkman.github.io

Education

2011–2015 **PhD Candidate**[1,2]

[1]Institute for Integrated and Intelligent Systems
Griffith University (Gold Coast, Australia)

[2]NICTA – National ICT Australia (Brisbane, Australia)

thesis: Predicting stability and functional changes induced by protein mutations with a machine learning approach

advisers: Prof Yaoqi Zhou[1], Prof Bela Stantic[1], Prof Abdul Sattar[1,2]

2007–2010 **Student of Informatics (Master's Degree)**

Faculty of Informatics
Masaryk University (Brno, Czech Republic)

2007 **Visiting Student**

School of Computer Science
University College Dublin (Dublin, Ireland)

2004–2007 **Student of Applied Informatics (Bachelor's Degree)**

Faculty of Informatics
Masaryk University (Brno, Czech Republic)

Work Experience

2018–now **Postdoctoral Researcher**

Medical Epigenomics Laboratory
CeMM Research Center for Molecular Medicine
Austrian Academy of Sciences (Vienna, Austria)

adviser: Prof Christoph Bock

2016–2018 **Postdoctoral Researcher**

Machine Learning & Computational Biology Lab
Department of Biosystems Science and Engineering
ETH Zurich (Basel, Switzerland)

adviser: Prof Karsten Borgwardt

2015–2016 **Visiting Postdoctoral Researcher**

Laboratory of Molecular Biology and Structural Bioinformatics
Institute for Integrated and Intelligent Systems
Griffith University (Gold Coast, Australia)

adviser: Prof Yaoqi Zhou

2008–2010 **Software Developer and Consultant**

ARTIN (Brno, Czech Republic)

Awards

Sep 2017	Online Poster Award (1st place) at [BC]^2 Basel Computational Biology Conference 2017 for 'Kernelized rank learning for personalized drug recommendation' (Basel, Switzerland)
Jul 2017	F1000Research Poster Award at ISMB/ECCB 2017 for 'Kernelized rank learning for personalized drug recommendation' (Prague, Czech Republic)
Jun 2014	Winner of Three-Minutes Thesis Competition (School of ICT, Griffith University)
Apr 2014	Gold Coast Association of Postgraduates Professional Development and Conference Funding Award
Jan 2013	Griffith University Conference Travel Grant
2011–2014	NICTA Project Award Scholarship

2011–2014	Griffith University Postgraduate Research Scholarship
2011–2014	Griffith University International Postgraduate Research Scholarship
Dec 2010	NICTA Summer Research Scholarship
Dec 2010	2nd best paper in the workshop Keymaker 2010 (Brno, Czech Republic)
May 2010	2nd place in a national research competition of Czech and Slovak university students 'SVOČ 2010' in the category of Applied Informatics (Ostrava, Czech Republic)
Mar 2010	Dean's award for outstanding thesis (Faculty of Informatics, Masaryk University)
Mar 2010	Dean's award for excellent study results (Faculty of Informatics, Masaryk University)
Sep–Dec 2007	Erasmus Student Exchange Scholarship

Publications

Journal articles:

1. He* X, **Folkman* L**, Borgwardt K: 'Kernelized rank learning for personalized drug recommendation', *Bioinformatics* (2018), 34(16), 2808–2816, DOI: 10.1093/bioinformatics/bty132
2. Livingstone* M, **Folkman* L**, Yang Y, Zhang P, Mort M, Cooper DN, Liu Y, Stantic B, Zhou Y: 'Investigating DNA-, RNA-, and protein-based features as a means to discriminate pathogenic synonymous variants', *Human Mutation* (2017), 38(10), 1336–1347, DOI: 10.1002/humu.23283.
3. **Folkman L**, Stantic B, Sattar A, Zhou Y: 'EASE-MM: sequence-based prediction of mutation-induced stability changes with feature-based multiple models', *Journal of Molecular Biology* (2016), 428(6), 1394–1405, DOI: 10.1016/j.jmb.2016.01.012.
4. **Folkman L**, Yang Y, Li Z, Stantic B, Sattar B, Mort M, Cooper DN, Liu Y, Zhou Y: 'DDIG-in: detecting disease-causing genetic variations due to frameshifting indels and nonsense mutations by sequence and structural properties at nucleotide and protein levels', *Bioinformatics* (2015) 31(10), 1599–1606, DOI: 10.1093/bioinformatics/btu862.
5. **Folkman L**, Stantic B, Sattar A: 'Feature-based multiple models improve classification of mutation-induced stability changes', *BMC Genomics* (2014) 15(Suppl 4), S6, DOI: 10.1186/1471-2164-15-S4-S6.
6. **Folkman L**, Stantic B, Sattar A: 'Towards sequence-based prediction of mutation-induced stability changes in unseen non-homologous proteins', *BMC Genomics* (2014) 15(Suppl 1), S4, DOI: 10.1186/1471-2164-15-S1-S4.
7. **Folkman L**, Stantic B, Sattar, A: 'Sequence-only evolutionary and predicted structural features for the prediction of stability changes in protein mutants', *BMC Bioinformatics* (2013) 14(Suppl 2), S6, DOI: 10.1186/1471-2105-14-S2-S6.

**** denotes equal contributions***

Conference proceedings:

8. Higgs T, **Folkman L**, Stantic B: Combining protein fragment feature-based resampling and local optimisation, *in* 'Pattern Recognition in Bioinformatics', Vol 7986 of *LNBI*, pp. 114–125, Springer, 2010.
9. **Folkman L**, Pullan W, Stantic B: Generic parallel genetic algorithm framework for protein optimisation, *in* 'Algorithms and architectures for parallel processing', Vol. 7017 of *LNCS*, pp. 64–73, Springer, 2011.
10. Stetsko A, **Folkman L**, Matyas V: Neighbor-based intrusion detection in wireless sensor networks, *in* 'Wireless and Mobile Communications (ICWMC), pp. 420–425, IEEE, 2010.

Teaching

2016	**Teaching Assistant** Department of Biosystems Science and Engineering, ETH Zurich *course:* Data Mining I
2012–2013	**Teaching Assistant** School of Information and Communication Technology, Griffith University *courses:* Introduction to Programming, Object-Oriented Programming, Database Systems and Administration, Database Technology

Skills

Languages	English – advanced (IELTS band score 8), German – beginner, Slovak – native
Programming	Python, Java, C, C++, SQL, Linux shell, parallel programming with MPI